# Networks and permissions

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## Accessibility Notice

This document is also available in HTML format at:

Link: `https://aholdengouveia.name/IntroLinux/labs/networking.html`

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

## Learning Outcomes

By the end of this lab, you will be able to:

- Use ping to test network connectivity
- View network settings using ip a or ifconfig
- Set file permissions using chmod
- Explain file permissions in your own words

## Lab Instructions

### Networking

You are going to use two basic commands: ping, and ifconfig or ip a. On Linux, ping keeps running until you stop it, so press Ctrl+C to stop it.

1. Run ping Google.com. Was ping successful? What was the result? Take a screenshot.
2. Run ping CNN.com. Was ping successful? What was the result? Take a screenshot.
3. Use ifconfig or ip a. What does it show you? Use man to see what other useful things one of those can do

### File Permissions

Create a subdirectory for this lab and put several files in it either using cp or touch

Use chmod to create files with the following permissions:

1. Everyone can read and write but not execute
2. Owner has full permissions but everyone else can only read
3. No one can do anything including the owner
4. Owner and group have read and write permission, public only has read
5. Owner can read and execute but not write, everyone else can read

Use ls -l to show results, take a screenshot and include in report

Pick one file and explain the permissions in your own words in the lab report

## Deliverables

One file with your name, date, and lab title at the top and all the answers.

- You should have screenshots for each ping, your ifconfig or ip a, and showing your file permissions
- Make sure to include labels for each screenshot so it's clear what you're turning in. Your screenshots must show your prompt with your username, or a note with your name and the term.
- Don't forget to include the answers to each question asked as well.

## Resources

References, a video, a PowerPoint and some notes are available at my website
Link: `https://www.aholdengouveia.name/IntroLinux/networking.html`

My FAQ Linux playlist on YouTube, with demos of many of the commands and scripts in this class Link: `https://www.youtube.com/playlist?list=PLQOmuLC45eoQlIMH7oCNL6yFP7kWQubBo`